

Comparative Analysis of i-RAN and O-RAN Architectures Enhanced by Quantum Optimization for 6G Wireless Networks

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ABSTRACT The continuous evolution towards sixth-generation (6G) wireless networks imposes stringent requirements on latency, reliability, scalability, and energy efficiency. To address these challenges, this paper presents a comprehensive and comparative study of four Radio Access Network (RAN) paradigms: the baseline intelligent RAN (i-RAN), its optimized variant leveraging Software Defined Networking (SDN) and Multi-access Edge Computing (MEC), the classical Open RAN (O-RAN) architecture enhanced by the RAN Intelligent Controller (RIC), and the proposed Quantum-Optimized O-RAN (Q-O-RAN), which integrates cutting-edge quantum technologies including Quantum Key Distribution (QKD), Quantum Blockchain (QBC), and Quantum Neural Networks (QNN). Through detailed simulation-based evaluations under realistic network conditions, we analyze key performance indicators such as average latency, system throughput, energy consumption, and end-to-end reliability. Results demonstrate that quantum-enhanced O-RAN significantly outperforms traditional models by reducing latency and energy usage while enhancing data rate and reliability. This work underscores the transformative potential of quantum intelligence in building secure, adaptive, and ultra-reliable 6G wireless infrastructures.

INDEX TERMS Intelligent RAN, Open RAN, Quantum Networks, QKD, QBC, QNN, 6G, SDN, MEC, Network Optimization

I. INTRODUCTION

The rapid explosion of mobile data traffic alongside the rise of latency-critical and mission-sensitive applications—such as holographic communication, autonomous vehicles, and immersive extended reality (XR)—demands a fundamental transformation in wireless network design. The evolution towards sixth-generation (6G) networks imposes stringent requirements including sub-millisecond latency, ultra-high reliability (99.999%), enhanced energy efficiency, and intelligent, context-aware decision-making within highly heterogeneous and dynamic environments.

Conventional Radio Access Networks (RANs), constrained by rigid architectures and centralized control, struggle to meet these escalating demands. To overcome these limitations, Intelligent RANs (i-RAN) have emerged, leveraging Software Defined Networking (SDN) and Multi-access Edge Computing (MEC) to enable greater flexibility, programmability, and localized management of network resources. Although i-RAN introduces intelligent and decentralized control elements, its performance remains constrained by tradi-

tional optimization techniques and bottlenecks in centralized decision-making.

Building on this foundation, Open RAN (O-RAN) further decentralizes and disaggregates RAN functions, employing open interfaces to allow interoperability among vendor-neutral components. The RAN Intelligent Controller (RIC), with its Near-Real-Time and Non-Real-Time modules, orchestrates AI-driven control and real-time policy enforcement across distributed network elements, fostering a scalable, adaptable, and interoperable ecosystem tailored for diverse 6G scenarios.

Parallel to these advances, quantum technologies present groundbreaking opportunities for network intelligence and security. Quantum Key Distribution (QKD) offers provably secure communication channels; Quantum Blockchain (QBC) enhances trust and immutability in distributed ledger systems; and Quantum Neural Networks (QNNs) enable exponential acceleration in learning and decision-making processes. Integrating these quantum capabilities into the O-RAN framework gives rise to Quantum-Optimized O-RAN

(Q-O-RAN), a next-generation architecture that synergizes classical AI with quantum advantages to fulfill the ultra-reliable low-latency communication (URLLC) demands of future networks.

In this study, we conduct a comprehensive analysis and performance comparison of four RAN paradigms: the baseline i-RAN with static resource allocation, the optimized i-RAN integrating SDN and MEC for dynamic control, the AI-driven O-RAN architecture with RIC-based orchestration, and the quantum-enhanced Q-O-RAN leveraging QKD, QBC, and QNN. Through realistic simulations encompassing diverse traffic profiles, user mobility, and service heterogeneity, we assess these models across key performance indicators such as latency, throughput, energy efficiency, and reliability.

Our results demonstrate that the quantum-optimized O-RAN significantly outperforms the other models, achieving notable improvements in latency reduction and security robustness, thereby positioning itself as a promising cornerstone for future 6G wireless systems.

The remainder of the paper is organized as follows: Section II presents the related work and motivation; Section III details the i-RAN system model with and without quantum enhancements; Section IV describes the O-RAN system with and without quantum enhancements; Section V provides simulation results and comparative analysis; finally, Section VI concludes and discusses future directions towards a fully quantum-enabled end-to-end 6G network.

II. RELATED WORK

The evolution of Radio Access Networks (RAN) has been a major research focus as networks progress from 5G toward the more demanding requirements of 6G. Initial research on intelligent RAN (i-RAN) architectures explored the integration of Software Defined Networking (SDN) and Multi-access Edge Computing (MEC) to enable flexible, context-aware, and low-latency communication infrastructures [22], [23]. These works demonstrated that decentralized computing at the network edge, combined with centralized SDN control, improves latency, bandwidth utilization, and scalability.

Parallel advancements in Open RAN (O-RAN) have promoted disaggregation and openness, fostering interoperability between hardware and software components. Standardized by the O-RAN Alliance [24], O-RAN introduces the RAN Intelligent Controller (RIC), composed of Near-Real-Time (Near-RT) and Non-Real-Time (Non-RT) modules, which facilitate closed-loop control, AI-driven decision-making, and enhanced orchestration [25]. Recent studies have leveraged deep reinforcement learning, federated learning, and graph neural networks within O-RAN to perform dynamic resource allocation, interference management, and network slicing optimization [26]–[28].

Simultaneously, quantum technologies are rapidly emerging as key enablers for the future of secure and intelligent communications. Quantum Key Distribution (QKD) enables provably secure key exchange mechanisms based on the laws of quantum mechanics, eliminating vulnerabilities inherent

in classical encryption schemes [13]. Quantum Blockchain (QBC) offers tamper-resistant ledgers with enhanced computational complexity and attack resilience [29]. Furthermore, Quantum Neural Networks (QNNs) provide new paradigms for data classification and decision-making, potentially surpassing the computational limits of classical neural networks in specific tasks [30], [31].

Emerging research efforts have begun to explore the integration of these quantum techniques within RAN infrastructures. For instance, Zhang et al. [32] propose a quantum-aware O-RAN framework combining QKD and QNN to enhance both security and inference efficiency in dynamic network environments. Moreover, recent works have investigated quantum-assisted resource allocation and scheduling algorithms that leverage QNNs for faster convergence and better generalization [33]. However, most current studies focus on theoretical feasibility or isolated use cases, lacking a comprehensive performance evaluation across classical and quantum-augmented RAN models.

In contrast, our work provides a unified and systematic comparative study of four distinct RAN architectures—i-RAN, optimized i-RAN (with SDN and MEC), classical O-RAN (with RIC), and quantum-enhanced O-RAN (Q-O-RAN). By simulating realistic deployment scenarios with heterogeneous traffic and mobility patterns, we aim to fill the research gap in evaluating how quantum enhancements influence key performance metrics such as latency, energy consumption, throughput, and system reliability.

III. MOTIVATION

The transition toward sixth-generation (6G) wireless networks introduces unprecedented performance requirements, including ultra-low latency—often targeting below 1 millisecond—extremely high reliability reaching up to 99.9999%, enhanced spectral efficiency, and robust end-to-end security [20], [22]. These stringent demands are propelled by cutting-edge applications such as autonomous vehicles, remote robotic surgery, industrial automation, and immersive augmented and virtual reality experiences, all of which require instantaneous, trustworthy, and intelligent communication infrastructures [10], [27].

While fifth-generation (5G) networks have laid the groundwork for intelligent and flexible connectivity, classical Radio Access Network (RAN) architectures face considerable challenges in scaling to the agility and context-awareness necessary for such diverse and dynamic use cases. Intelligent RAN (i-RAN) models have emerged by integrating Software-Defined Networking (SDN) and Multi-access Edge Computing (MEC), enabling programmable control and distributed intelligence at the network edge [5], [18]. These enhancements facilitate improved resource management and responsiveness; however, real-time decision-making remains constrained by increasing user mobility, heterogeneous traffic patterns, and evolving security threats [26].

The advent of Open RAN (O-RAN) marks a significant architectural shift by emphasizing openness, vendor-neutral

interfaces, and modular design. The introduction of RAN Intelligent Controllers (RICs) empowers O-RAN with AI-driven real-time optimization capabilities, facilitating more adaptive and fine-grained resource orchestration [24], [25]. Nonetheless, the openness and decentralization intrinsic to O-RAN also raise critical concerns regarding network security, privacy preservation, and trustworthiness in distributed decision processes [32].

In response to these challenges, quantum technologies present promising avenues to enhance both the security and intelligence of next-generation networks. Quantum Key Distribution (QKD) provides provably secure key exchange protocols based on the principles of quantum mechanics, guaranteeing information-theoretic security immune to both classical and quantum computational attacks [13], [15]. Quantum Blockchain (QBC) leverages quantum principles to bolster data integrity and immutability across distributed ledgers, enhancing trust and transparency in network resource management and transaction validation [14], [29]. Moreover, Quantum Neural Networks (QNN) offer powerful models capable of processing high-dimensional data and complex spatiotemporal patterns with potential exponential speed-ups, making them ideal candidates for advanced network intelligence tasks such as anomaly detection, traffic prediction, and dynamic resource allocation [11], [16], [31].

This study is motivated by the convergence of these emerging quantum technologies with evolving RAN paradigms. By systematically comparing classical i-RAN and O-RAN architectures with their quantum-augmented counterparts, we aim to elucidate the tangible benefits that quantum-assisted intelligence and security can bring. In particular, we examine how the integration of SDN and MEC in i-RAN enhances flexibility [5]; how RIC-enabled O-RAN architectures improve orchestration and AI-driven decision-making [25]; and how quantum-optimized O-RAN (Q-O-RAN) architectures achieve transformative improvements in latency, energy efficiency, and security [32], [33]. Through this comprehensive analysis, our objective is to provide a clear roadmap for the design of future-proof, secure, and intelligent wireless infrastructure that meets the demanding requirements of 6G and beyond.

IV. I-RAN WITH AND WITHOUT QUANTUM OPTIMIZATION

The Intelligent Radio Access Network (i-RAN) considered in this work includes a set of N gNodeBs connected to Multi-access Edge Computing (MEC) servers, orchestrated by Software-Defined Networking (SDN) controllers. The RAN Intelligent Controller (RIC) plays a central role in traffic management, resource allocation, and network optimization. We analyze both the classical i-RAN architecture and its quantum-enhanced version (Quantum-i-RAN), which integrates Quantum Neural Networks (QNN), Quantum Key Distribution (QKD), and Quantum Blockchain (QBC).

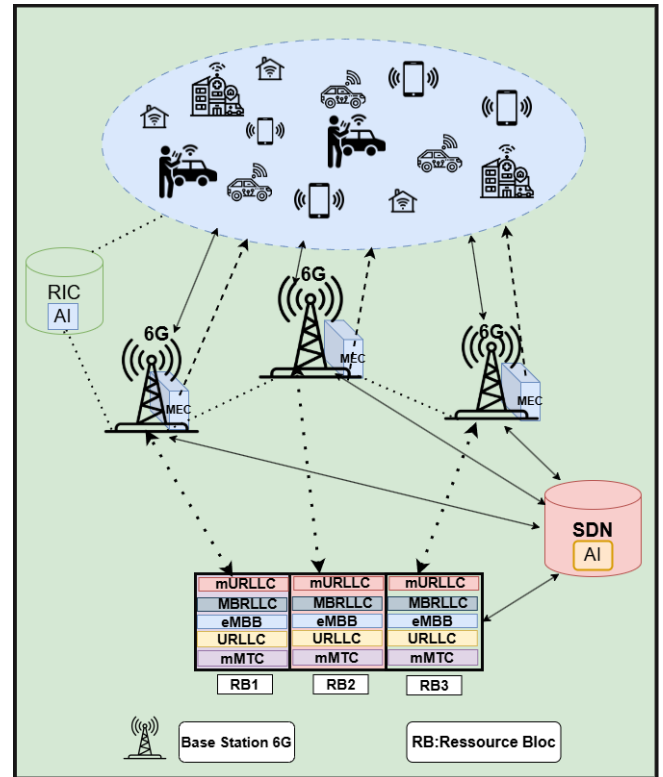


FIGURE 1. i-RAN Architecture

1) System Model of Classical i-RAN

The Intelligent Radio Access Network (i-RAN) is built upon a centralized and intelligent architecture designed to optimize radio resource management in modern mobile networks. This system consists of several key components: a set of gNodeBs providing radio connectivity, Multi-access Edge Computing (MEC) servers that bring processing capabilities closer to end users, a central controller known as the RAN Intelligent Controller (RIC), and a network control infrastructure based on Software-Defined Networking (SDN). Artificial intelligence is integrated primarily within the RIC to enable adaptive and predictive decision-making. In the i-RAN architecture (Figure 1), the RIC functions as a centralized controller responsible for orchestrating dynamic radio resource management, load balancing, and quality of service optimization. Unlike the O-RAN architecture, where the RIC is split into Near-Real-Time and Non-Real-Time components, the i-RAN RIC consolidates all control and optimization functions, leveraging advanced AI algorithms to analyze real-time data collected from the network.

The gNodeBs maintain the radio connection with user terminals and continuously report key performance metrics such as traffic load, channel conditions, and resource usage to the RIC. Using this information, the RIC drives the allocation of radio resources and coordinates the gNodeBs' actions to maximize network efficiency.

The SDN controller complements the RIC by providing

flexible and programmable control over the network's control plane, enabling dynamic routing and flow reconfiguration.

MEC servers deliver decentralized processing capabilities, executing critical functions locally—such as traffic forecasting and localized optimization—to reduce latency and alleviate the load on the central network.

By embedding AI within both the RIC and potentially at the MEC level, the i-RAN system operates autonomously, continuously adapting network parameters based on traffic fluctuations, signal quality, and service requirements (e.g., eMBB, URLLC, mMTC).

Overall, the i-RAN architecture presents a centralized and intelligent infrastructure where the RIC serves as the pivotal controller, orchestrating radio resources, coordinating gNodeBs, leveraging MEC capabilities, and managing the network via SDN to ensure optimal quality of service.

2) Analytical Study of Classical i-RAN

a: Latency Model:

Assuming an M/M/1 queuing model for each gNodeB, the average latency is:

$$L_i(t) = \frac{1}{\mu_i - \lambda_i(t)} \quad (1)$$

where $\lambda_i(t)$ is the packet arrival rate at gNodeB i and μ_i is its service rate.

b: Energy Model:

The energy consumption at gNodeB i is given by:

$$E_i(t) = E_{\text{idle}} + \alpha_i \cdot a_i(t) \quad (2)$$

where E_{idle} is the base energy consumption, and α_i is a proportionality constant.

c: Reliability Model:

The reliability of gNodeB i is modeled as:

$$R_i(t) = e^{-\beta l_i(t)} \quad (3)$$

with β being an interference sensitivity parameter.

d: Utility Function:

The overall system utility is a weighted combination of the key performance metrics:

$$U(t) = \sum_{i=1}^N \left[w_1 \frac{1}{L_i(t)} + w_2 T_i(t) - w_3 E_i(t) + w_4 R_i(t) \right] \quad (4)$$

where $T_i(t)$ is the throughput and w_j are operator-defined weights.

3) System Model of Quantum-i-RAN (Q-iRAN)

The Quantum-i-RAN (Q-iRAN) architecture (Figure 2) builds upon the classical i-RAN framework by integrating cutting-edge quantum technologies to enhance intelligence, security, and trustworthiness at critical network layers. This integration aims to meet the stringent demands of next-generation 6G networks through improved performance, robustness, and security.

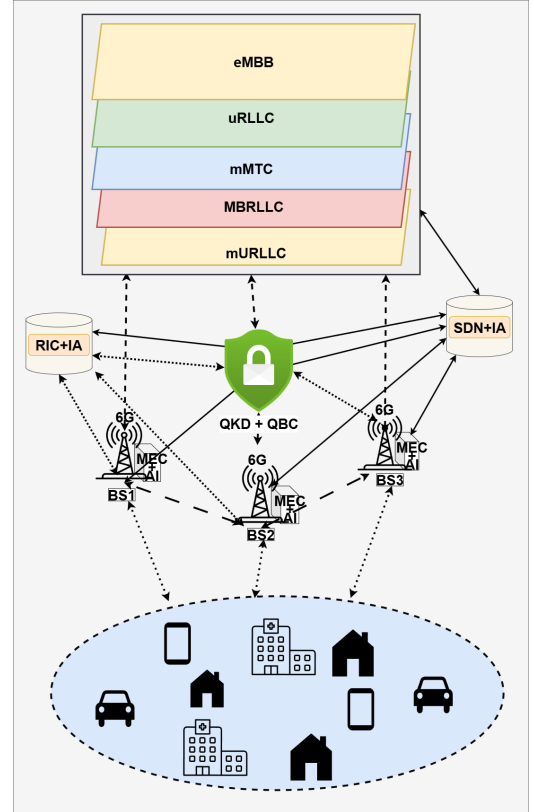


FIGURE 2. Q-iRAN Architecture

Central to Q-iRAN is the deployment of **Quantum Neural Networks (QNN)**, which empower the RAN Intelligent Controller (RIC) with superior computational capabilities for traffic prediction and dynamic resource allocation. Leveraging quantum superposition and entanglement, QNNs provide faster inference and greater model expressiveness compared to classical AI methods, enabling real-time adaptive optimization under complex and volatile network conditions.

Security is fortified through **Quantum Key Distribution (QKD)**, which guarantees unconditional security for cryptographic key exchange between network entities such as gNodeBs, MEC servers, and the RIC. By exploiting the principles of quantum mechanics, QKD detects eavesdropping attempts and ensures that encryption keys remain secret, thus safeguarding communication against both classical and quantum adversaries.

To ensure data integrity and transparency in resource management, Q-iRAN employs **Quantum Blockchain (QBC)**

technology. QBC enhances traditional blockchain frameworks by integrating quantum-resistant cryptographic functions and quantum consensus mechanisms. This results in an immutable distributed ledger that prevents tampering and ensures reliable transaction records, crucial for secure and auditable network operations.

Together, these quantum technologies transform the i-RAN into a more intelligent, secure, and trustworthy system capable of addressing the multifaceted challenges posed by ultra-dense, high-throughput, and latency-sensitive 6G scenarios.

4) Analytical Study of Quantum-i-RAN

a: QNN-based Resource Allocation:

The QNN model is a parameterized quantum circuit $Q(\theta)$, learning the mapping:

$$f : \mathbf{s}(t) \mapsto \mathbf{a}(t)$$

The learning task minimizes a quantum loss function:

$$\min_{\theta} \mathcal{L}(\theta) = \mathbb{E}_{\mathbf{s}(t)} [\|\mathbf{a}(t) - Q(\mathbf{s}(t); \theta)\|^2] \quad (5)$$

where $Q(\mathbf{s}(t); \theta) = \langle \psi(\mathbf{s}(t), \theta) | M | \psi(\mathbf{s}(t), \theta) \rangle$ represents the circuit output with measurement M .

b: QKD: Secure Key Exchange via BB84 Protocol

In BB84 [15], each bit is encoded as a qubit in one of two bases:

$$\mathcal{B}_Z = \{|0\rangle, |1\rangle\}, \quad \mathcal{B}_X = \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\}$$

After basis reconciliation and error checking, the final key rate R satisfies:

$$R \geq n [1 - H_2(e) - f_{EC} H_2(e)] \quad (6)$$

where $n \approx N/2$, e is the QBER, f_{EC} is the error correction efficiency, and $H_2(e)$ is the binary entropy.

Security is guaranteed by:

$$\Pr[\text{key compromised}] \leq \epsilon \quad (7)$$

c: QBC: Quantum-Resistant Blockchain

Quantum blockchains utilize quantum-resistant hash functions H_q and entangled state-based consensus mechanisms. Hash collision probability is:

$$\Pr[H_q(x) = H_q(y), x \neq y] \approx 0 \quad (8)$$

ensuring data immutability and increasing $R_i(t)$.

5) Impact of Quantum Integration on Performance Metrics

Quantum-i-RAN modifies the classical performance models as follows:

a: Latency:

$$L_i^Q(t) = \gamma_L \cdot L_i(t), \quad 0 < \gamma_L < 1 \quad (9)$$

b: Energy Consumption:

$$E_i^Q(t) = \gamma_E \cdot E_i(t), \quad 0 < \gamma_E < 1 \quad (10)$$

c: Reliability:

$$R_i^Q(t) = R_i(t) + \Delta R, \quad \Delta R > 0 \quad (11)$$

d: Throughput:

$$T_i^Q(t) = T_i(t) + \Delta T, \quad \Delta T > 0 \quad (12)$$

6) Conclusion

The Quantum-i-RAN framework leverages quantum computing and communication techniques to significantly improve the performance of next-generation wireless networks. By integrating QNN, QKD, and QBC, the system achieves lower latency, enhanced security, better energy efficiency, and higher throughput—aligning with the ambitious requirements of 6G and beyond.

A. PROPOSED ALGORITHM

To efficiently manage radio resources in both classical and quantum-optimized i-RAN architectures, we propose a dynamic resource allocation algorithm driven by intelligent decision-making at the RAN Intelligent Controller (RIC). The algorithm leverages real-time network state information, predictive analytics, and adaptive control to optimize throughput, latency, energy consumption, and reliability.

a: Algorithm Overview

The proposed algorithm operates iteratively in discrete time intervals, executing the following key steps at each iteration:

- 1) **Network State Collection:** Gather up-to-date metrics from all gNodeBs, including traffic load, channel quality indicators, and energy consumption statistics.
- 2) **Traffic Prediction and Load Estimation:** Employ advanced AI models—classical neural networks in i-RAN or Quantum Neural Networks (QNN) in Quantum-i-RAN—to forecast short-term traffic demands and load fluctuations.
- 3) **Resource Optimization:** Based on the predicted traffic, solve an optimization problem to allocate radio resources $\mathbf{a}(t)$ across gNodeBs, balancing trade-offs among latency, throughput, energy efficiency, and reliability, while respecting capacity and QoS constraints.
- 4) **Security Enforcement:** For Quantum-i-RAN, ensure secure key exchange through Quantum Key Distribution (QKD) and maintain tamper-proof records via Quantum Blockchain (QBC).
- 5) **Resource Allocation and Control:** Dispatch allocation decisions to the gNodeBs and MEC servers for scheduling and local optimization.
- 6) **Feedback and Adaptation:** Continuously monitor network performance, updating AI models and refining resource allocation dynamically.

b: Algorithmic Framework

Our approach employs a hybrid Deep Reinforcement Learning (DRL) agent based on Proximal Policy Optimization (PPO), enhanced with a Graph Neural Network (GNN) to exploit network topology and spatial correlations among gNodeBs and user equipments (UEs). In the quantum-optimized scenario (Quantum-i-RAN), the classical GNN is replaced or complemented by a Quantum Neural Network (QNN), leveraging quantum superposition and entanglement for richer state representations and faster convergence.

Algorithm 1 PPO-GNN/QNN-Based Resource Allocation for i-RAN

Initialize DRL agent with policy π_θ and value function V_ϕ . Construct graph $\mathcal{G}$ from network topology (gNodeBs and UEs) network is operational. Observe current state s_t (channel conditions, traffic demands, QoS). Encode s_t using GNN (or QNN in Quantum-i-RAN). Sample action $a_t \sim \pi_\theta(s_t)$ for resource allocation. Execute action a_t and observe reward r_t and next state s_{t+1} . Update parameters θ and ϕ to maximize PPO clipped objective. Quantum-i-RAN Perform Quantum Key Distribution (QKD) key exchanges. Update Quantum Blockchain (QBC) ledger.

c: Mathematical Objective

The PPO agent optimizes the policy parameters θ by maximizing the clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio, $\hat{A}_t$ the advantage estimate, and ϵ the clipping parameter.

This adaptive algorithm framework empowers the i-RAN architecture to efficiently respond to rapidly changing network conditions by learning optimal resource allocation policies. The integration of quantum-enhanced QNNs in Quantum-i-RAN accelerates convergence and improves decision-making quality, positioning the system for next-generation 6G network challenges.

V. O-RAN WITH AND WITHOUT QUANTUM OPTIMIZATION

A. SYSTEM MODEL: O-RAN WITH AND WITHOUT QUANTUM OPTIMIZATION

The proposed system adopts a modular and disaggregated Open Radio Access Network (O-RAN) architecture (Figure 3), designed to meet the performance, flexibility, and intelligence demands of next-generation 6G networks. O-RAN enables interoperability between components through open interfaces, and integrates advanced AI/ML algorithms for intelligent and automated management. The O-RAN architecture is composed of the following three physical units:

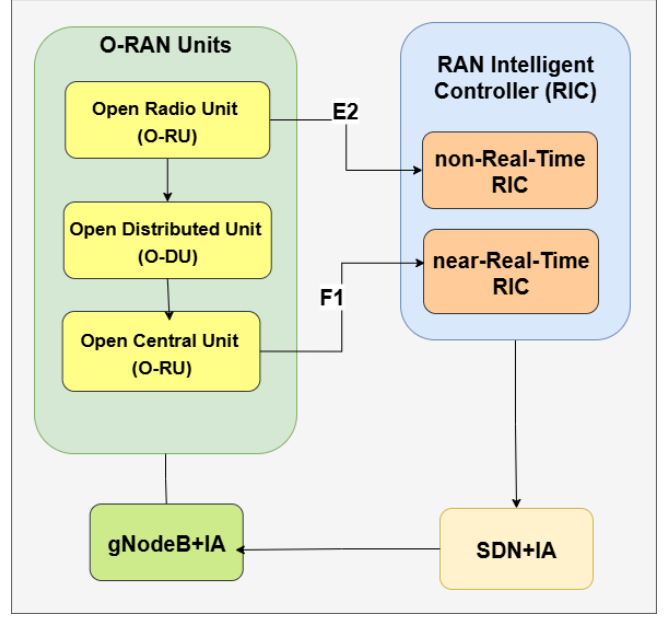


FIGURE 3. O-RAN Architecture

- **O-RU (Radio Unit):** Handles RF signal transmission and reception at the edge of the network.
- **O-DU (Distributed Unit):** Responsible for time-critical lower-layer protocols and real-time baseband processing.
- **O-CU (Central Unit):** Manages higher-layer protocol processing and user-plane control, operating in a non-real-time domain.

In addition to these physical components, the architecture includes a centralized intelligent controller:

- **RAN Intelligent Controller (RIC):** Divided into two logical layers:
 - **Near-Real-Time RIC (Near-RT RIC):** Executes control functions at sub-second granularity (10ms–1s) through xApps to dynamically manage radio resources and optimize network KPIs.
 - **Non-Real-Time RIC (Non-RT RIC):** Operates on timescales above 1s to perform AI model training, policy generation, and long-term analytics via rApps.

Classical O-RAN Model: In the baseline architecture, resource management and decision-making are handled using conventional deep reinforcement learning (DRL) methods. In particular, the *Proximal Policy Optimization (PPO)* algorithm is employed to learn optimal policies. To handle the non-Euclidean structure of the network topology (e.g., user-gNodeB relationships), the model integrates *Graph Neural Networks (GNNs)* for spatial embedding and representation learning.

Quantum-Augmented O-RAN (Q-O-RAN): To enhance the learning efficiency and representation power of classical models, we introduce quantum computing modules into the

AI pipeline. Specifically:

- **Quantum Neural Networks (QNNs):** Replace traditional deep learning layers with *variational quantum circuits (VQCs)* that exploit quantum entanglement and superposition to capture complex feature dependencies in high-dimensional data.
- **Quantum Error Correction (QEC):** Maintains computational integrity by mitigating noise and decoherence in quantum circuits, essential under Noisy Intermediate-Scale Quantum (NISQ) hardware.
- **Quantum Key Distribution (QKD):** Ensures secure key exchange across O-RAN components using quantum cryptographic protocols such as BB84 [15], making the network resilient to quantum-based eavesdropping.

Let the system consist of N gNodeBs $\{g_1, g_2, \dots, g_N\}$ serving a set of users $U = \{u_1, u_2, \dots, u_M\}$ over a shared spectrum resource pool R . Each gNodeB g_i has a dynamic bandwidth $B_i(t)$ available at time t , and each user u connected to g_i has a data demand $D_{i,u}(t)$. These parameters are processed by the RIC to orchestrate resources and make intelligent allocation decisions in either a classical or quantum-augmented context.

B. ANALYTICAL STUDY

The objective is to maximize network performance through intelligent resource allocation. Let $\mathcal{U}_i$ denote the set of users connected to gNodeB g_i . The total utility to be maximized can be modeled as:

$$\max_{\pi} \sum_{i=1}^N \sum_{u \in \mathcal{U}_i} U(R_{i,u}(t), D_{i,u}(t)), \quad (13)$$

where $R_{i,u}(t)$ represents the spectrum resource allocated to user u at time t , and $U(\cdot)$ is a utility function representing performance trade-offs (e.g., throughput, delay, and fairness). For instance, proportional fairness is captured via $U(R, D) = \log(R/D)$.

a: Classical Case (O-RAN without Quantum Optimization):

In this scenario, the policy $\pi^{\text{classical}}$ is learned using a PPO algorithm enhanced with a GNN encoder. The system state s_t encodes real-time parameters such as user mobility, channel conditions, traffic type, and gNodeB load. The policy is formulated as:

$$\pi^{\text{classical}}(s_t) = \text{PPO-GNN}(s_t; \theta), \quad (14)$$

where θ represents the trainable parameters of the neural network model. The GNN maps the system state into a latent representation, which is then used by PPO to select actions (resource allocation decisions).

b: Quantum-Enhanced Case (Q-O-RAN):

In the quantum-augmented system, the classical policy is extended by integrating a Quantum Neural Network (QNN) module. The GNN embeddings are passed to a variational

quantum circuit parameterized by rotation angles and entanglement gates:

$$\pi^{\text{quantum}}(s_t) = \text{QNN-PPO-GNN}(s_t; \theta, \phi), \quad (15)$$

where ϕ represents the quantum circuit parameters optimized through hybrid quantum-classical training (e.g., parameter-shift rule). This model captures higher-order correlations between spatial and temporal features, improving learning in complex network scenarios.

The integration of QNNs enables better generalization in high-dimensional and stochastic environments, while QEC ensures resilience against quantum noise. Moreover, the use of QKD ensures that RIC-gNodeB communications remain secure against both classical and quantum attacks.

C. PROPOSED ALGORITHM: QUANTUM-ASSISTED INTELLIGENT O-RAN

Algorithm 2 Quantum-Assisted PPO-GNN for O-RAN Resource Allocation

- 1: **Input:** Network graph $G = (V, E)$, user demand matrix $D_{i,u}$, spectrum pool R
- 2: **Initialize:** Classical parameters θ , quantum parameters ϕ
- 3: **for** each training episode **do**
- 4: Observe system state s_t : includes user mobility, CSI, interference, and demand
- 5: Generate graph embeddings: $h_t = \text{GNN}(s_t)$
- 6: Evaluate value function via QNN: $Q^\phi(h_t)$
- 7: Select action $a_t = \pi(h_t)$ using PPO with QNN-enhanced critic
- 8: Apply a_t , observe reward r_t and next state s_{t+1}
- 9: Update θ, ϕ using hybrid gradient optimization
- 10: **end for**
- 11: **Output:** Trained policy π^* for optimal resource allocation

This hybrid quantum-classical approach provides a scalable, secure, and adaptive framework for managing complex 6G network environments. The synergy of GNN (for topology-aware encoding), PPO (for robust policy optimization), and QNN (for efficient value estimation) enables Q-O-RAN to outperform traditional models, particularly under high user density, dynamic traffic, and adversarial security conditions.

VI. PERFORMANCE EVALUATION

A. SIMULATION ENVIRONMENT

The simulation experiments were conducted on the Google Colab platform, which offers a versatile and accessible cloud-based environment well-suited for the development and evaluation of sophisticated machine learning and reinforcement learning models. Leveraging Colab's powerful computational resources, including GPUs and TPUs, we were able to efficiently train both classical deep neural networks and their quantum-enhanced counterparts without requiring local high-performance hardware.

The platform's seamless integration with popular Python libraries such as Stable Baselines3 and OpenAI Gym enabled rapid prototyping, iterative experimentation, and effective benchmarking of diverse Radio Access Network (RAN) architectures. Furthermore, Colab's collaborative features ensured easy sharing and reproducibility of simulation results, facilitating transparent and open research practices.

This cloud-based setup significantly accelerated the development cycle of the intelligent RAN (i-RAN) models, enabling scalable simulation of complex scenarios representative of next-generation 6G wireless networks.

B. SIMULATION PARAMETERS AND MODELING APPROACH

Our simulation framework was designed to closely emulate a realistic 6G network environment, encompassing multiple base stations (gNodeBs), Multi-access Edge Computing (MEC) servers, and centralized control entities such as Software-Defined Networking (SDN) controllers and the RAN Intelligent Controller (RIC). These components collaborate to orchestrate network resources intelligently and dynamically.

The simulation scenario integrates heterogeneous traffic types aligned with 6G service categories, including enhanced Mobile Broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive Machine-Type Communications (mMTC). Dynamic variations in network load and traffic patterns were incorporated to mimic realistic network conditions, user mobility, and stress-testing of resource allocation strategies.

Key simulation parameters, summarized in Table 2, include the number of gNodeBs, MEC servers, controllers, and the duration of each simulation episode.

TABLE 1. Key Simulation Parameters

Parameter	Value
Number of gNodeBs	3
Number of MEC Servers	2
Number of SDN Controllers	1
Number of RICs	1
Simulation Steps per Episode	100
Traffic Types	eMBB, URLLC, mMTC

At the core of the simulation is a reinforcement learning environment modeled using the OpenAI Gym framework, with multiple variants representing the classical i-RAN, optimized i-RAN, O-RAN, and Quantum-enhanced O-RAN (Q-O-RAN) architectures. Each environment simulates key performance metrics such as latency, throughput, energy consumption, and reliability, incorporating stochastic elements to reflect network dynamics.

Quantum-specific components were abstractly modeled to capture the benefits of Quantum Neural Networks (QNN) in accelerated inference, Quantum Key Distribution (QKD) for secure communication, and Quantum Blockchain (QBC)

for trustworthy resource management. The simulation also includes simplified cryptographic verification and ledger consistency checks to emulate security assurances in quantum-enhanced scenarios.

Training and evaluation were performed using the Proximal Policy Optimization (PPO) algorithm from the Stable Baselines3 library. Each model was trained over multiple episodes with continuous monitoring of performance indicators, enabling comprehensive comparative analysis of the architectures under realistic 6G conditions.

This integrated simulation approach provides a robust and scalable framework to assess the potential benefits and trade-offs of integrating quantum technologies into intelligent RAN architectures, thereby guiding future research and practical deployments.

C. SIMULATION PARAMETERS AND MODELING APPROACH

Our simulation framework was designed to closely emulate a realistic 6G network environment, encompassing multiple base stations (gNodeBs), Multi-access Edge Computing (MEC) servers, and centralized control entities such as Software-Defined Networking (SDN) controllers and the RAN Intelligent Controller (RIC). These components collaborate to orchestrate network resources intelligently and dynamically.

The simulation scenario integrates heterogeneous traffic types aligned with 6G service categories, including enhanced Mobile Broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive Machine-Type Communications (mMTC). Dynamic variations in network load and traffic patterns were incorporated to mimic realistic network conditions, user mobility, and stress-testing of resource allocation strategies.

Key simulation parameters, summarized in Table 2, include the number of gNodeBs, MEC servers, controllers, and the duration of each simulation episode.

TABLE 2. Key Simulation Parameters

Parameter	Value
Number of gNodeBs	3
Number of MEC Servers	2
Number of SDN Controllers	1
Number of RICs	1
Simulation Steps per Episode	100
Traffic Types	eMBB, URLLC, mMTC

At the core of the simulation is a reinforcement learning environment modeled using the OpenAI Gym framework, with multiple variants representing the classical i-RAN, optimized i-RAN, O-RAN, and Quantum-enhanced O-RAN (Q-O-RAN) architectures. Each environment simulates key performance metrics such as latency, throughput, energy consumption, and reliability, incorporating stochastic elements to reflect network dynamics.

Quantum-specific components were abstractly modeled to capture the benefits of Quantum Neural Networks (QNN) in accelerated inference, Quantum Key Distribution (QKD) for secure communication, and Quantum Blockchain (QBC) for trustworthy resource management. The simulation also includes simplified cryptographic verification and ledger consistency checks to emulate security assurances in quantum-enhanced scenarios.

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This integrated simulation approach provides a robust and scalable framework to assess the potential benefits and trade-offs of integrating quantum technologies into intelligent RAN architectures, thereby guiding future research and practical deployments.

D. RESULTS AND INTERPRETATION

This section presents a detailed comparative analysis of the four architectures under investigation: i-RAN, optimized i-RAN (Quantum-i-RAN), O-RAN, and optimized O-RAN (Quantum-O-RAN). The evaluation is based on four critical performance metrics: latency, throughput, energy consumption, and reliability.

E. COMPARISON BETWEEN I-RAN AND QUANTUM-I-RAN

This subsection presents a detailed comparison between the traditional i-RAN architecture and its quantum-optimized counterpart, Quantum-i-RAN, which integrates Quantum Neural Networks (QNN), Quantum Key Distribution (QKD), and Quantum Blockchain (QBC). The evaluation is based on four critical performance indicators: latency, throughput, energy consumption, and reliability.

Figure 4 illustrates the latency evolution over time for both architectures. The results show a significant latency reduction of approximately 40% in the Quantum-i-RAN, decreasing from 53.26 ms to 31.98 ms. This improvement is attributed to the enhanced inference capabilities of the QNN, which accelerates the decision-making process at the gNodeB level.

Regarding energy consumption, Figure 5 shows a modest improvement of 4.7%, with the optimized model consuming 5.41 J compared to 5.68 J for the baseline. This gain results from the more efficient orchestration of computational and communication tasks enabled by the quantum mechanisms.

On the other hand, Figure 6 presents a slight decrease in throughput, dropping from 57.03 Mbps to 53.43 Mbps (a reduction of 6.3%). This decline is mainly due to the cryptographic overhead introduced by QKD and the distributed verification inherent to QBC. While this introduces marginal delay in data transmission, it is a necessary trade-off to achieve higher security guarantees and integrity.

Lastly, as shown in Figure 7, the reliability remains unchanged at a high level of 0.95 in both cases, confirming that

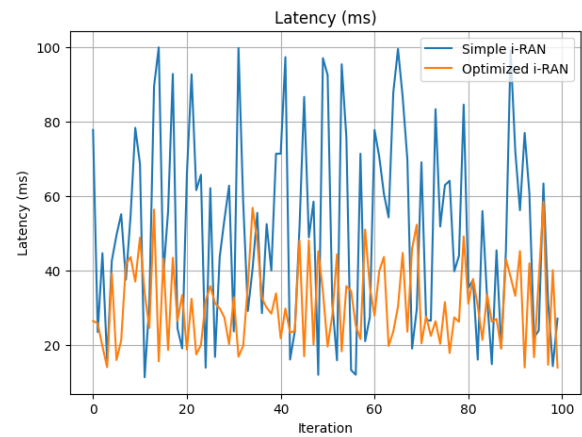


FIGURE 4. Latency comparison between i-RAN and Quantum-i-RAN

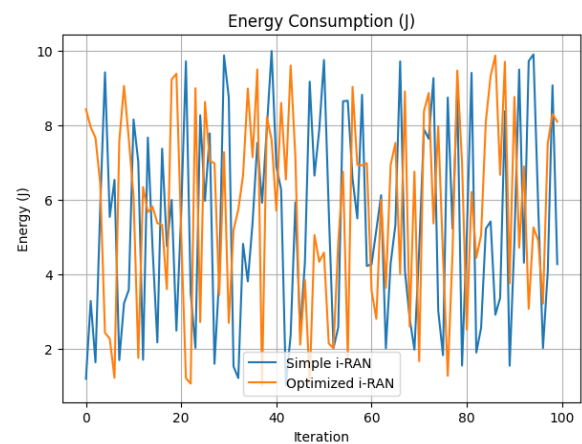


FIGURE 5. Energy consumption comparison between i-RAN and Quantum-i-RAN

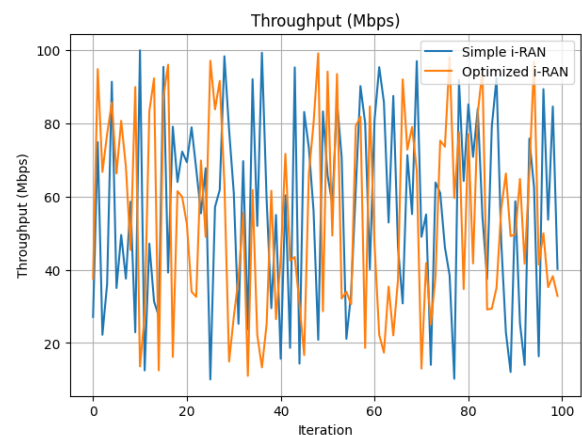


FIGURE 6. Throughput comparison between i-RAN and Quantum-i-RAN

the integration of quantum technologies does not compromise the stability or consistency of the network.

Overall, the Quantum-i-RAN architecture demonstrates significant improvements in latency and energy efficiency,

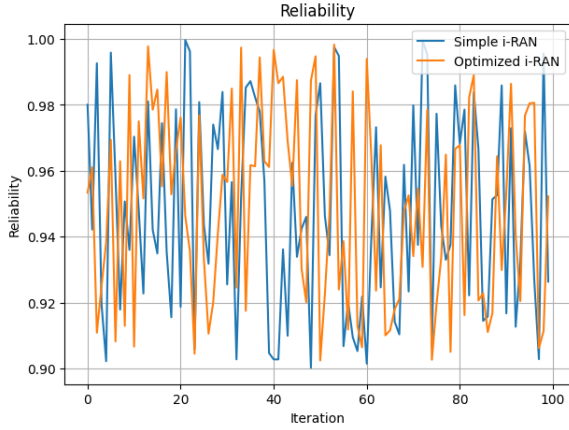


FIGURE 7. Reliability comparison between i-RAN and Quantum-i-RAN

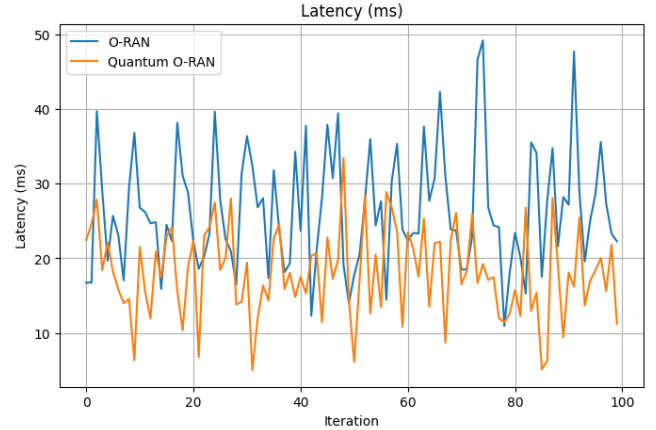


FIGURE 8. Latency comparison between O-RAN and Quantum-O-RAN

while slightly sacrificing throughput to reinforce data integrity and security. This balance exemplifies the practical value of quantum enhancements in future 6G systems. The performance comparison is summarized in Table 3, which clearly outlines the advantages of the Quantum-i-RAN over its classical counterpart.

TABLE 3. Performance Metrics of i-RAN and Quantum-i-RAN

Model	Latency (ms)	Throughput (Mbps)	Energy (J)	Reliability
i-RAN	53.26	57.03	5.68	0.95
Quantum-i-RAN	31.98	53.43	5.41	0.95

F. COMPARISON BETWEEN O-RAN AND QUANTUM-O-RAN

The integration of quantum technologies into the classical Open RAN (O-RAN) architecture has led to the development of the **Quantum-O-RAN** model. This enhanced version leverages the open, modular principles of O-RAN while incorporating cutting-edge quantum components such as Quantum Neural Networks (QNN), Quantum Key Distribution (QKD), and Quantum Blockchain (QBC). The goal of this integration is to address the increasingly stringent requirements of 6G networks, particularly in terms of low latency, high reliability, energy efficiency, and data security.

A detailed comparison between the two architectures is presented in Table 4, showcasing significant performance improvements across all key metrics. Firstly, the latency is dramatically reduced by approximately **38%**, decreasing from 27.94 ms in classical O-RAN to just 17.37 ms in Quantum-O-RAN. This substantial gain is primarily attributed to the superior inference capabilities of QNNs embedded in the RAN Intelligent Controller (RIC), which facilitate faster and more accurate decision-making in resource allocation processes.

In terms of throughput, Quantum-O-RAN achieves a noticeable improvement, rising from 70.10 Mbps to **80.52 Mbps**, an increase of nearly **15%**. This boost results from

the intelligent orchestration and dynamic adaptation provided by quantum-enhanced control logic, which optimizes spectrum utilization and minimizes transmission congestion under varying network conditions.

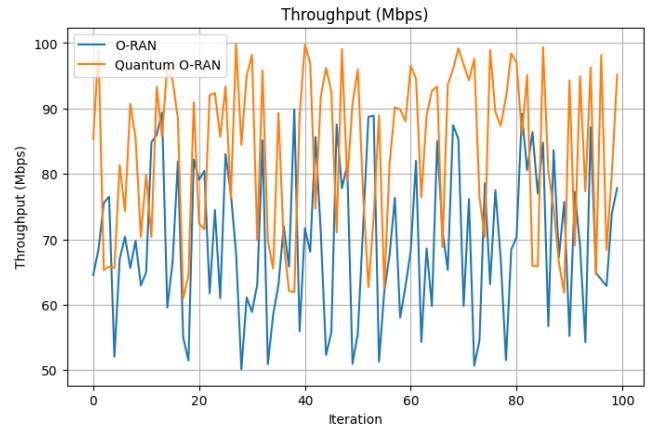


FIGURE 9. Throughput comparison between O-RAN and Quantum-O-RAN

Energy efficiency is also significantly enhanced. The average energy consumption drops from 4.09 J to **3.49 J**, representing a **15%** reduction. This improvement is a direct outcome of the intelligent and adaptive scheduling enabled by quantum optimization techniques, which reduce unnecessary transmissions and improve processing efficiency.

Lastly, the reliability metric increases from 0.96 to a remarkable **0.98**, highlighting the strengthened robustness and trustworthiness of the system. This enhancement is enabled by the use of QKD for secure and tamper-proof key exchange and QBC to ensure transparency, immutability, and integrity in control transactions and resource policies.

These advancements collectively confirm the strong potential of Quantum-O-RAN in meeting the demands of next-generation wireless systems. By combining the openness and interoperability of O-RAN with the computational and cryptographic superiority of quantum technologies, Quantum-O-

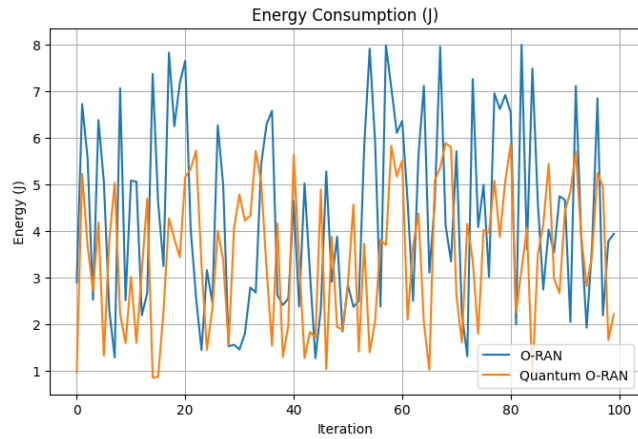


FIGURE 10. Energy consumption comparison between O-RAN and Quantum-O-RAN

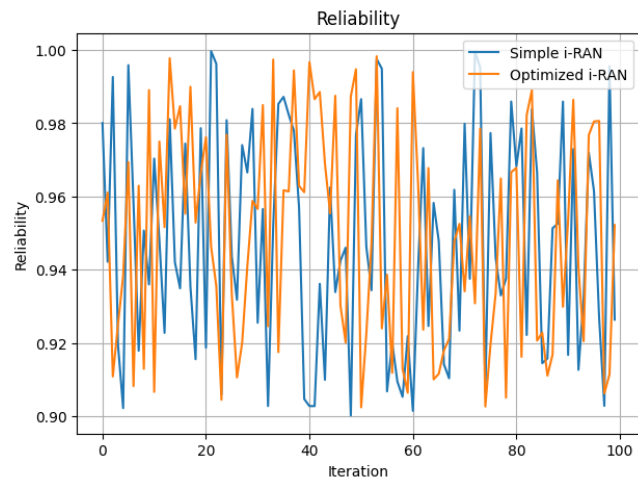


FIGURE 11. Reliability comparison between O-RAN and Quantum-O-RAN

RAN emerges as a compelling architecture for 6G deployments, especially in mission-critical and latency-sensitive applications.

TABLE 4. Performance Metrics of O-RAN and Quantum-O-RAN

Model	Latency (ms)	Throughput (Mbps)	Energy (J)	Reliability
O-RAN	27.94	70.10	4.09	0.96
Quantum-O-RAN	17.37	80.52	3.49	0.98

G. OVERALL COMPARATIVE ANALYSIS

Table 5 provides a consolidated summary of the key performance indicators across the four architectures considered: classical i-RAN, Quantum-i-RAN, O-RAN, and Quantum-O-RAN. The results clearly demonstrate the outstanding performance of the Quantum-O-RAN model, which achieves the lowest latency, highest throughput, most efficient energy

consumption, and greatest reliability among all architectures.

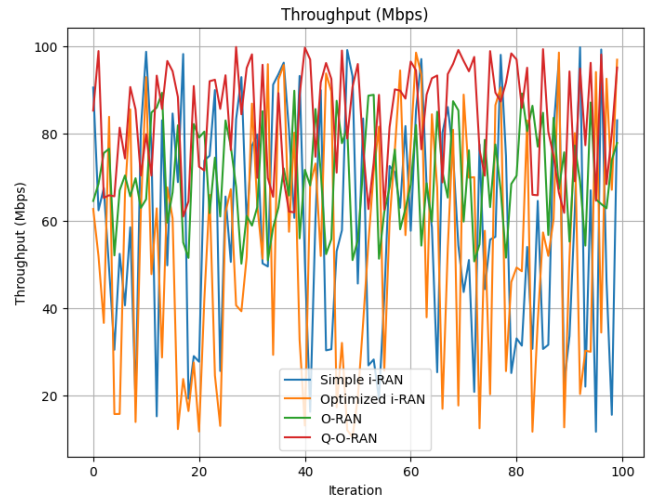


FIGURE 12. Throughput comparison across all architectures

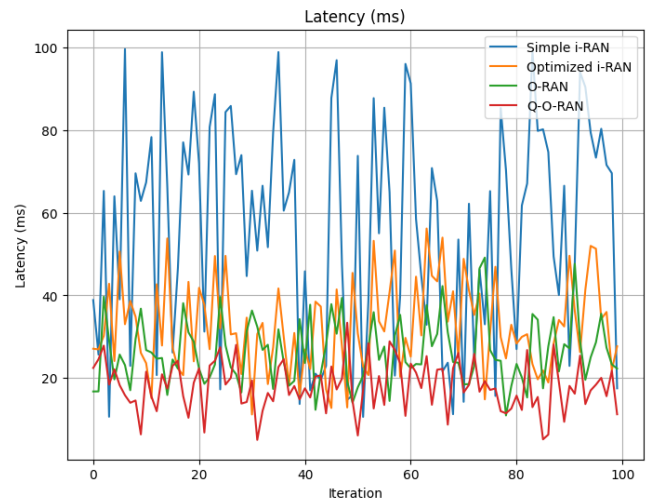


FIGURE 13. Latency comparison across all architectures

The classical i-RAN exhibits the highest latency and energy consumption, alongside the lowest throughput. These results highlight the inherent limitations of legacy architectures in fully meeting the stringent requirements of future 6G networks, particularly in terms of speed and energy efficiency. Although the incorporation of quantum technologies in Quantum-i-RAN substantially reduces latency and energy usage, a slight decrease in throughput is observed, primarily due to the overhead introduced by quantum security protocols.

In contrast, the O-RAN architecture benefits from its open, modular, and programmable design, resulting in significant performance improvements over classical i-RAN. Its quantum-enhanced counterpart, Quantum-O-RAN, further capitalizes on Quantum Neural Networks (QNN) for

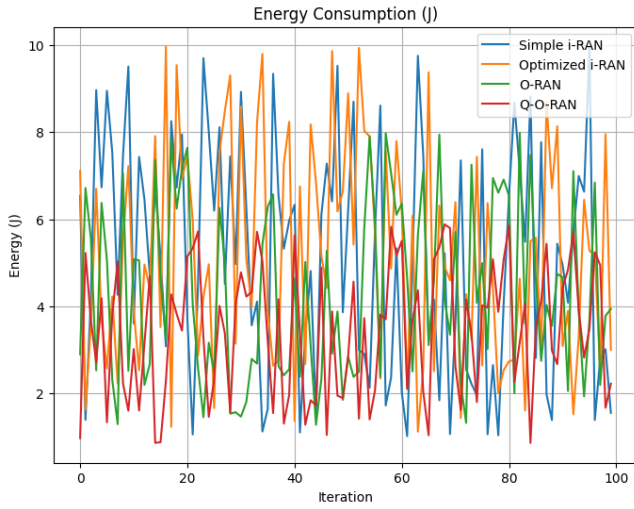


FIGURE 14. Energy consumption comparison across all architectures

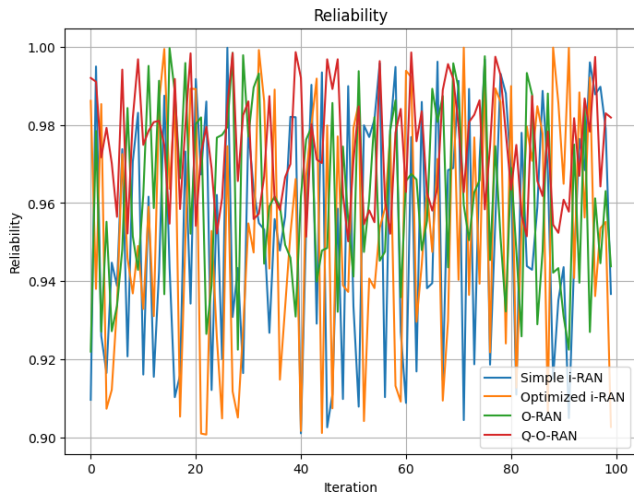


FIGURE 15. Reliability comparison across all architectures

accelerated processing, Quantum Key Distribution (QKD) for robust security, and Quantum Blockchain (QBC) for trustworthy resource management. This integration leads to pronounced gains in throughput and reliability while simultaneously reducing latency and energy consumption — all critical factors for sustainable and secure 6G deployments.

a: Conclusion

These findings underscore the essential value of integrating quantum technologies such as QNN, QKD, and QBC into advanced radio access network frameworks. Quantum-O-RAN emerges as a highly promising architecture capable of delivering low-latency, high-throughput, energy-efficient, and secure wireless communications. This study highlights the pivotal role of quantum-assisted intelligence in shaping next-generation open and intelligent RAN architectures.

TABLE 5. Overall Performance Comparison of Studied Architectures

Architecture	Latency (ms)	Throughput (Mbps)	Energy (J)	Reliability
i-RAN	53.26	57.03	5.68	0.95
Quantum-i-RAN	31.98	53.43	5.41	0.95
O-RAN	27.94	70.10	4.09	0.96
Quantum-O-RAN	17.37	80.52	3.49	0.98

VII. CONCLUSION AND PERSPECTIVES

This study has thoroughly examined the transformative impact of integrating quantum technologies—specifically Quantum Neural Networks (QNN), Quantum Key Distribution (QKD), and Quantum Blockchain (QBC)—within advanced Radio Access Network (RAN) architectures. Through rigorous simulations combined with reinforcement learning-based performance evaluations, we contrasted the classical i-RAN and O-RAN models against their quantum-enhanced counterparts, revealing substantial improvements in key metrics.

The findings decisively demonstrate that the Quantum-O-RAN architecture significantly outperforms all other considered frameworks by delivering markedly lower latency, enhanced throughput, reduced energy consumption, and superior reliability. These performance gains arise from the harmonious interplay between quantum-assisted inference accelerating dynamic resource allocation, quantum cryptographic protocols ensuring unbreakable data security, and distributed quantum ledger technologies guaranteeing transparency and trustworthiness across the network. Although Quantum-i-RAN achieves notable latency and energy efficiency improvements, it experiences a slight throughput reduction attributable to the overhead imposed by quantum security mechanisms.

The results affirm that the convergence of quantum technologies with open, intelligent RAN paradigms represents a highly promising avenue for next-generation 6G wireless networks. Such integration facilitates ultra-reliable, low-latency, energy-efficient, and secure communications, which are essential to meet the increasingly stringent demands of emerging applications ranging from autonomous systems to immersive extended reality.

Looking ahead, several compelling research directions emerge. Foremost, the deployment and validation of these quantum-empowered RAN architectures on practical testbeds and real-world environments will be vital to evaluate scalability, interoperability, and resilience under dynamic conditions. Further investigation into advanced quantum neural network architectures, incorporating quantum error correction and noise mitigation strategies, promises to enhance inference accuracy and robustness. Moreover, synergizing quantum-enhanced RANs with cutting-edge paradigms such as edge computing, network slicing, and AI-driven orchestration could unlock unprecedented levels of adaptability, resource optimization, and network intelligence. Additionally,

comprehensive exploration of quantum-resistant cross-layer security and privacy protocols will be critical to safeguard 6G networks against evolving cyber threats and ensure user data confidentiality.

In summary, this work establishes foundational insights into the quantum-driven evolution of wireless access networks and paves the way for realizing intelligent, secure, and sustainable 6G communications ecosystems. The integration of quantum technologies into open RAN frameworks is poised to redefine the future of wireless connectivity by enabling unprecedented performance, security, and flexibility.

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